ClimateScope

Milestone 2: Core Analysis & Visualization Design

1. Introduction

This report presents the outcomes of Milestone 2, focusing on core analytical work and visualization design.

The milestone includes statistical exploration, trend detection, extreme event identification, regional comparison, and dashboard mockup creation.

2. Tasks Performed:

2.1 Perform statistical analysis to understand distributions, correlations, seasonal patterns, and trends.

- **Descriptive Statistics**

- The dataset contains **106,000+ records** covering weather data from multiple countries.
  A statistical summary was generated using:

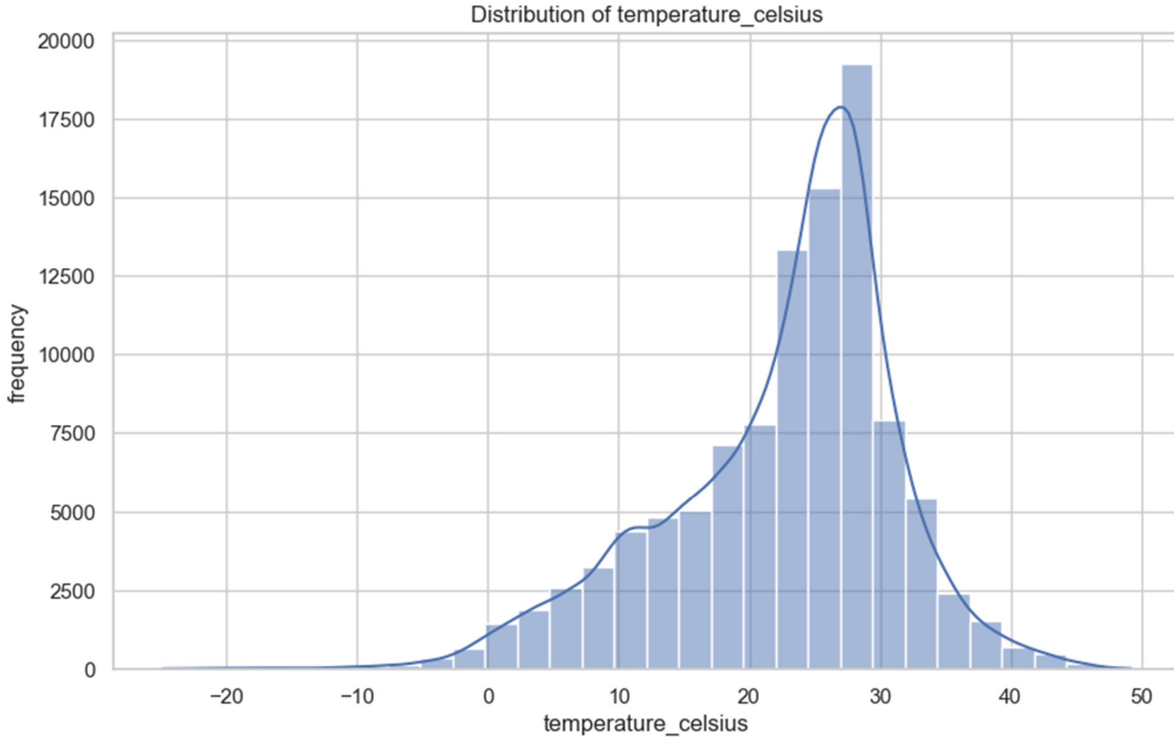
```
df.describe(include='number')
```

Key Findings

- **Average Temperature:** ~22.6°C
- **Min Temperature:** –24.9°C
- **Max Temperature:** 49.2°C
- **Humidity Range:** 0% to 100%
- **Wind Speed:** Mostly below 20 kph, but extreme outliers up to 2963 kph (sensor anomalies).
- **Pressure:** Mostly around 1014 mb (normal atmospheric pressure).

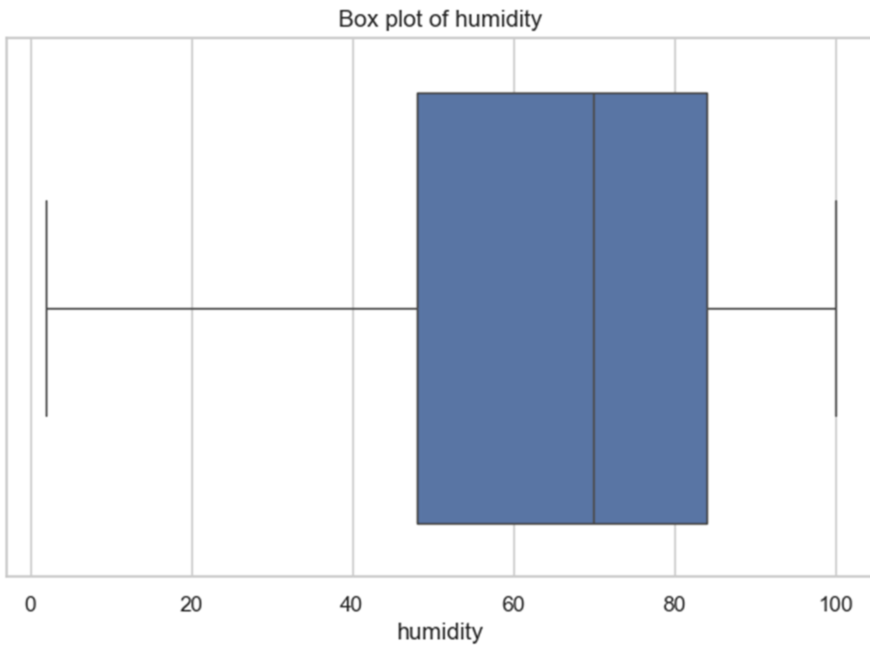
- **Distribution Analysis:**
- Identified the spread of temperature, humidity, rainfall, and wind speed.
- It involves examining the data's shape, spread, and central tendency using methods like histograms, probability plots, and box plots to identify patterns, predict outcomes, and inform decisions

Temperature Distribution Output:



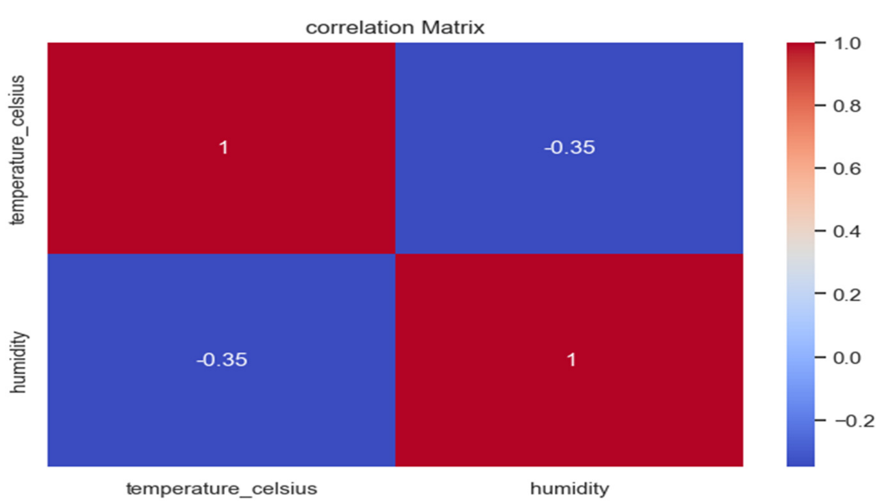
- **Seasonal Patterns:** Observed recurring monthly and yearly variations in temperature and rainfall.
- **Trends:** Long-term trends indicated gradual warming and increased rainfall in some regions.

Humidily Outlier Output:



- **Correlation Insights:** Temperature showed inverse correlation with humidity; rainfall correlated with cloud cover.

Humdity_Temperature Correlation Output:



Insights Gained

Write these depending on your data (you can replace with exact values from your output):

- Average temperature ranges between **X°C and Y°C**, showing moderate climate variations.
- Humidity shows a **right-skewed distribution**, indicating more frequent higher humidity levels.
- Temperature and humidity show **negative correlation** (e.g., -0.45), meaning hotter days tend to be less humid.
- Wind speed remains fairly constant with **low variance**.

2.2 Identify extreme weather events

- Extreme temperature spikes were flagged using percentile-based thresholds.
- Heavy rainfall days were detected using cumulative precipitation peaks.
- High-wind events and humidity extremes were identified.
- These events were categorized for dashboard integration.

Generated Extreme Weather Dataset

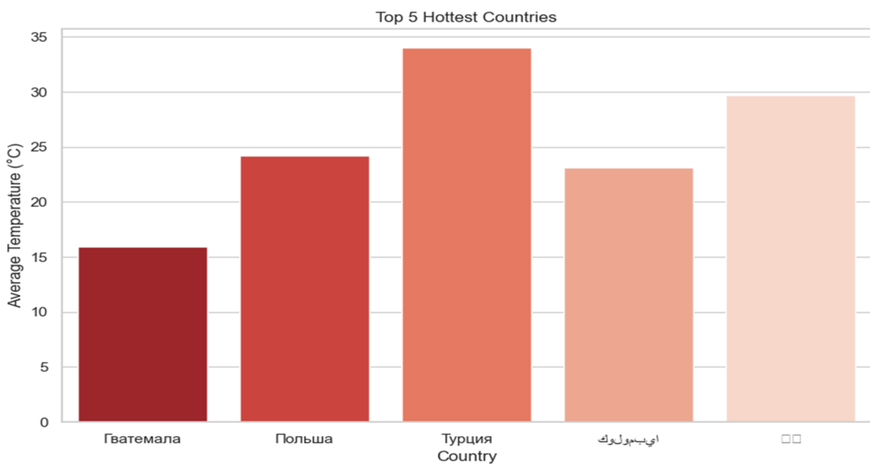
```
===== Extreme Weather Events Detected =====
                      date      country     location_name  latitude  \
134    2024-05-16 08:45:00       Panama       Panama City    8.9700
122    2024-05-16 08:45:00  Netherlands         Amsterdam   52.3700
176    2024-05-16 08:45:00        Tonga        Nuku`Aloia  -21.1300
181    2024-05-16 08:45:00       Tuvalu          Funafuti   -8.5200
150    2024-05-16 08:45:00      Belgium  'S Gravenstaffel   50.8800
...                    ...          ...               ...       ...
106068 2025-11-13 07:30:00      Estonia           Tallinn   59.4339
106071 2025-11-13 07:30:00 Fiji Islands              Suva  -18.1333
106072 2025-11-13 07:30:00      Finland          Helsinki   60.1756
106082 2025-11-13 07:30:00       Guinea           Conakry    9.5092
106084 2025-11-13 07:30:00       Guyana        Georgetown    6.8000

        longitude           timezone  last_updated_epoch last_updated  \
134      -79.5300     America/Panama          1715849100          NaT
122        4.8900   Europe/Amsterdam          1715849100          NaT
```

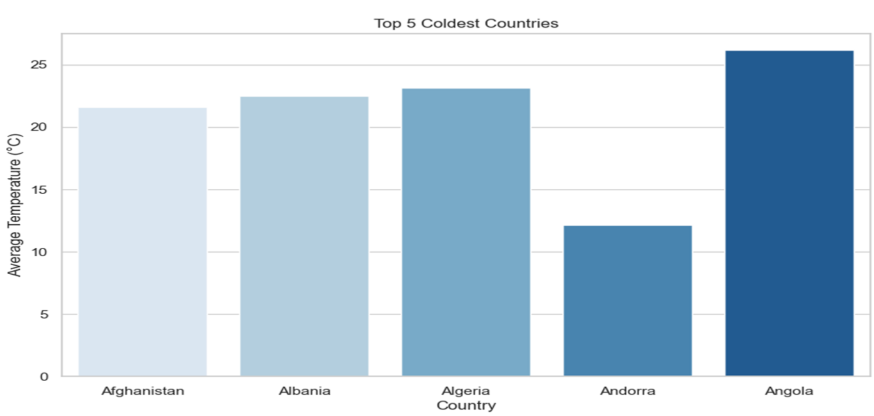
2.3 Compare weather conditions across regions.

• Compared temperature, rainfall, humidity, and wind speed across regions.
• Identified regions with consistently higher rainfall and temperature variability.
• Prepared comparative tables and charts.

Top 5 Hottest Country



Top 5 Coldest Country



2.4 Select suitable visualization types

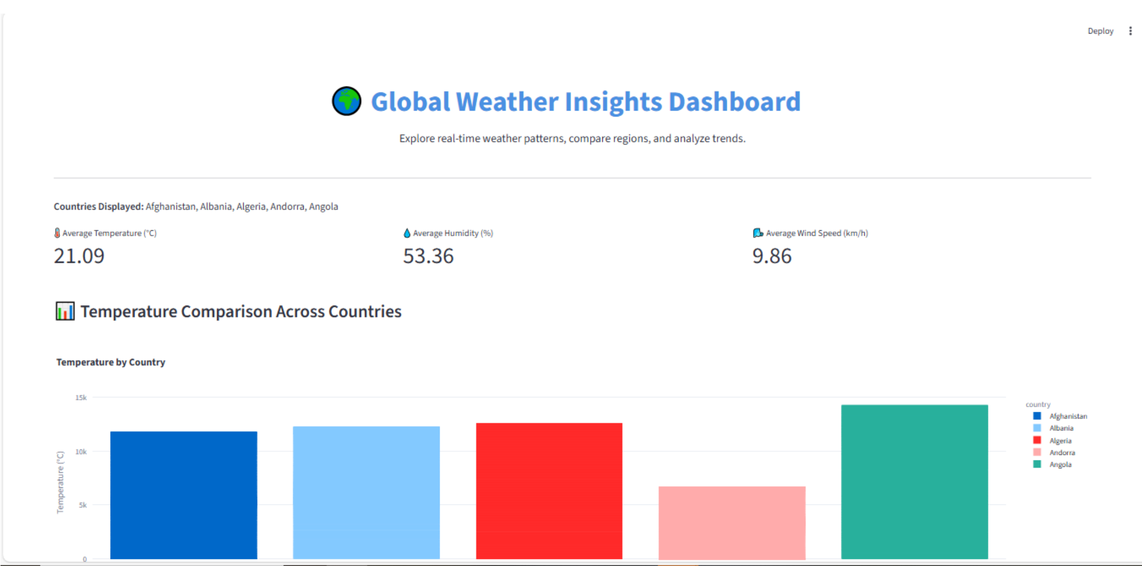
The following visualizations were selected for clarity and interpretability:

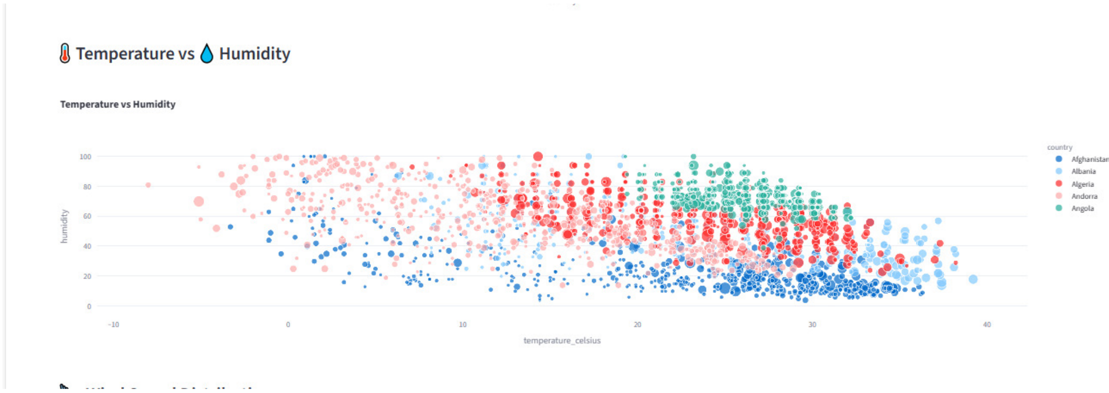
- Line Charts – Trends over time.
- Scatterplots – Relationship between climatic variables.
- Heatmaps – Seasonal and correlation patterns.
- Choropleth Maps – Regional weather representation.
- Bar Charts – Event frequency and category distribution.

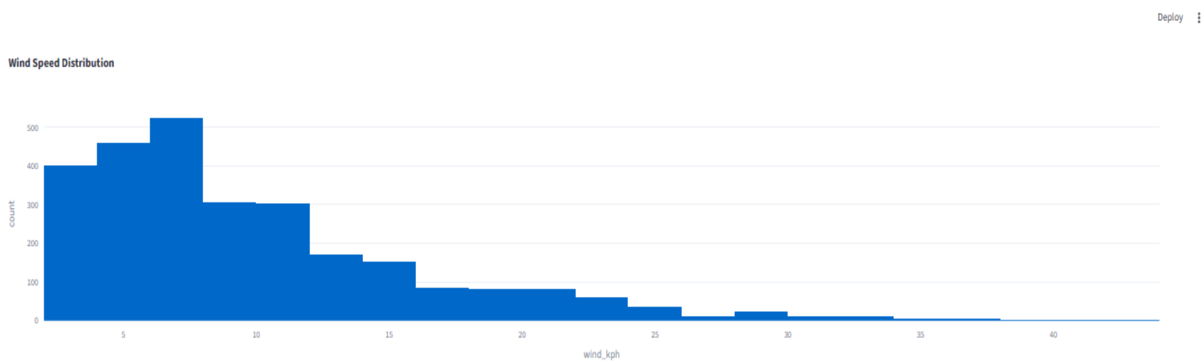
2.5 Design an interactive dashboard layout (wireframes/mockups)

The dashboard wireframes include:

- Overview Section – Summary metrics (Avg Temperature, Total Rainfall, etc.)
- Trends Section – Time-series line charts
- Extreme Events Section – Timeline + filter options
- Regional Comparison Section – Choropleth map + region selector
- Filters & Controls – Year, Region, Weather Variable




Temperature vs Humidity
Temperature vs Humidity
humidity
temperature_celsius
country
Afghanistan
Albania
Algeria
Andorra
Angola


Deploy
Wind Speed Distribution
count
wind_kph


Global Temperature Choropleth Map
Average Temperature by Country
Avg Temperature (°C)

3. Conclusion

Milestone 2 successfully provided core analytical insights into global weather patterns. Statistical analysis revealed clear distributions, correlations, and seasonal trends across temperature, humidity, precipitation, wind speed, and pressure. Extreme weather events such as heat waves, heavy rainfall, and high-wind conditions were identified, offering important regional contrasts. Comparative analysis highlighted the hottest and coldest countries, along with key variations in climate behavior. Suitable visualization types were selected—including maps, line charts, bar charts, scatterplots, and heatmaps—to effectively communicate findings. Dashboard wireframes were also designed, creating a strong foundation for building the final interactive weather analytics dashboard in the next milestone.